



Quantum Circuit Optimization

Why need Optimization for quantum circuit?

Each gate operation introduces noise.

Reduce gate size reduce the error rate for the circuit



Qiskit/Tket... Fixed sequence of passes and heuristic

Queso/Guoco/Quartz

Auto-synthesize thousands of rewrite rules

Apply exploration/superoptimization algorithm

Search-Based Methods:

Heuristics-Based Methods:

Prinwerk

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Prior work

Heuristics-Based Methods:

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Existing Problem

Heuristics-Based Methods:

Qiskit/Tket... Fixed sequence of passes and heuristic

Not effective for arbitrary circuits; no guarantee for coverage

Search-Based Methods:

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Apply exploration/superoptimization algorithm

Large set of rules; limited expressiveness; limited symbolic rewrite